Zambia, step 2 (2026-11-02)

chain	seed	iterations.saved	acceptance.rate	final.log.likelihood	best.log.likelihood	errored.proposals	file
1	20263016	5000	0.277	−38679.3	−38663.5	0	mcmc_zambia_step2_chain1_2026-10-31.Rdata
2	20263017	5000	0.277	−38690.2	−38674.3	0	mcmc_zambia_step2_chain2_2026-10-31.Rdata
3	20263018	5000	0.278	−38584.6	−38575.0	0	mcmc_zambia_step2_chain3_2026-11-01.Rdata
4	20263019	5000	0.275	−38685.8	−38678.2	0	mcmc_zambia_step2_chain4_2026-11-01.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):

age.20.to.29.transmission.multiplier.0	male.hiv.mortality.multiplier.0	age.15.to.19.transmission.multiplier.0
2.387	2.191	2.080
male.awareness.multiplier	age.25.to.49.hiv.mortality.multiplier.0	hiv.specific.mortality.rates.1
2.075	2.020	1.994
age.15.to.19.transmission.multiplier.3	male.hiv.mortality.multiplier.1	hiv.specific.mortality.rates.0
1.906	1.764	1.731
age.assortativity	age.20.to.29.transmission.multiplier.2	trate.2
1.725	1.682	1.625
age.30.to.39.transmission.multiplier.3	age.20.to.29.transmission.multiplier.1	trate.0
1.620	1.583	1.573
hiv.specific.mortality.rates.2	age.50.and.over.transmission.multiplier.3	trate.1
1.492	1.466	1.456
age.30.to.39.transmission.multiplier.0	age.15.to.19.transmission.multiplier.1	age.15.to.24.hiv.mortality.multiplier.2
1.456	1.456	1.402
age.20.to.29.transmission.multiplier.3	male.suppression.multiplier	age.0.to.14.hiv.mortality.multiplier.1
1.397	1.395	1.390
log.OR.testing.intercept		
1.369		

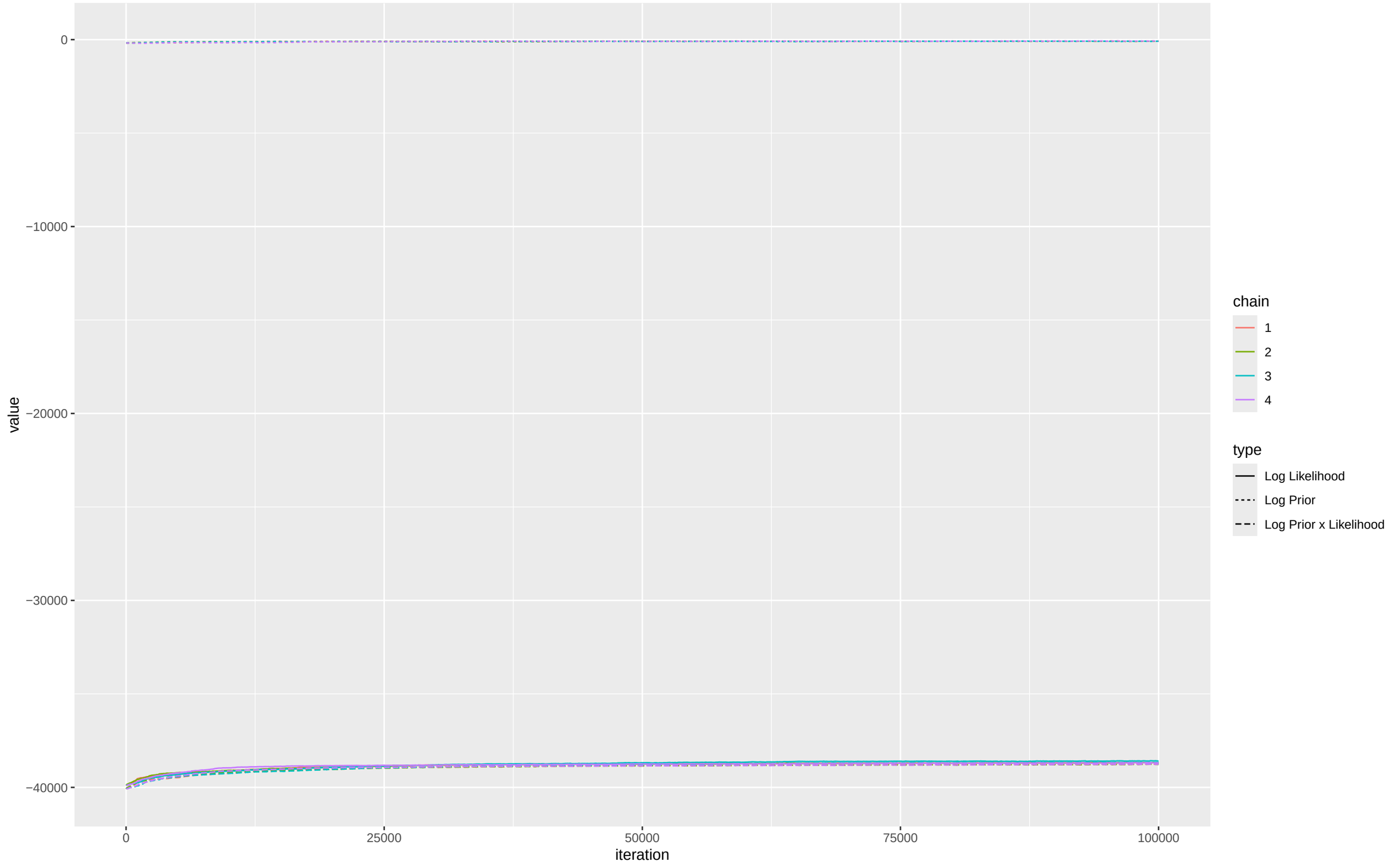
Merged simset:

location	simulations	total.weight	file
Zambia	1000	1000	simset_zambia_step2_chains1234_2026-11-01.Rdata

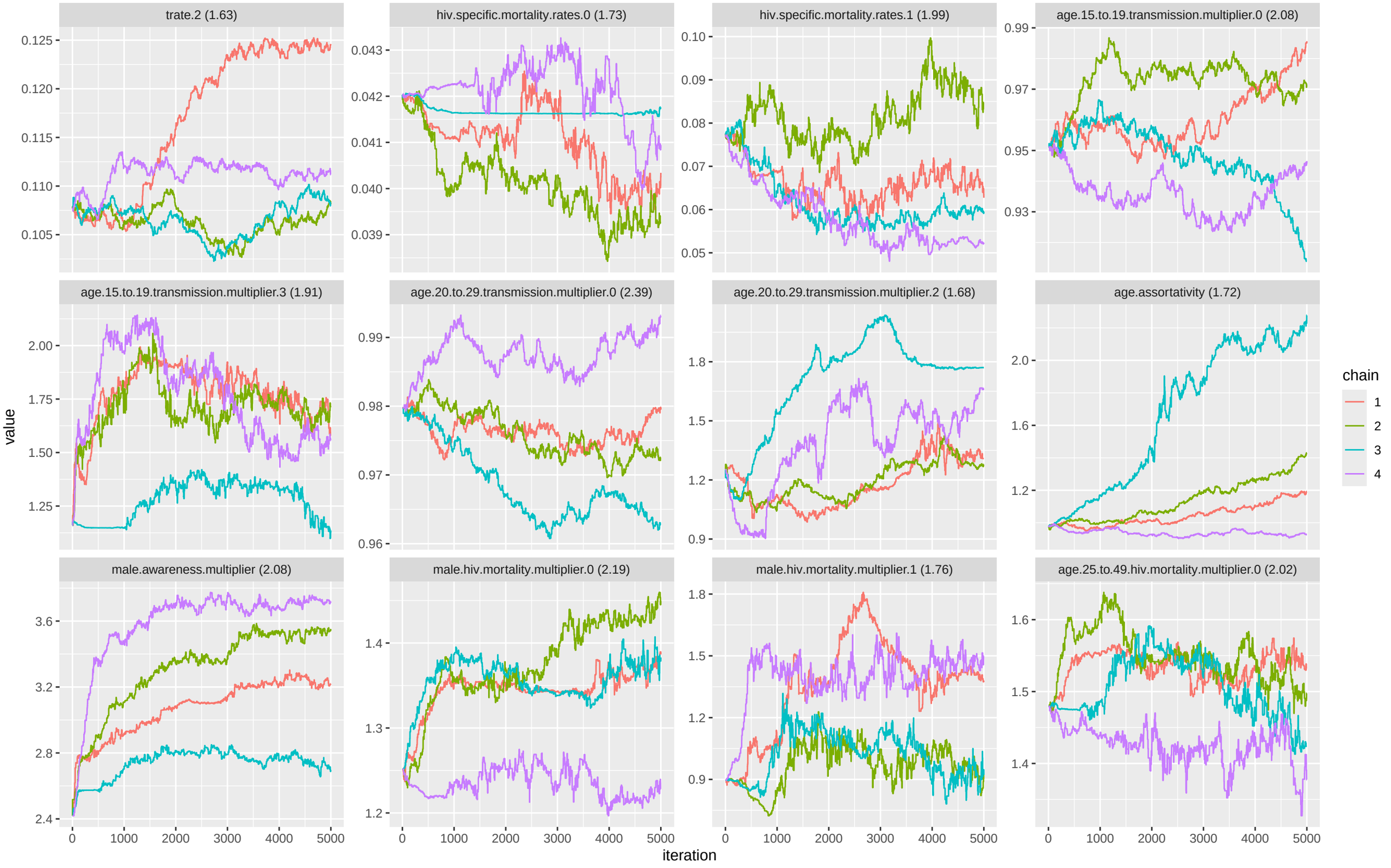
Fit check (merged simset vs every calibration target, 2000-2025): 57 of 63 targets flagged (worst 30 shown; all in <location>_fit_check.csv)

data.type	age	sex	years.with.data	pct.years.in.sim.95	median.sim.vs.data	last.year	last.sim.vs.data
hiv.mortality	25-49	female	26	0	x0.12	2025	x0.02
hiv.mortality	25-49	both	26	0	x0.20	2025	x0.02
hiv.mortality	25-49	male	26	0	x0.32	2025	x0.02
hiv.mortality	50 and over	female	26	0	x3.02	2025	x3.04
hiv.mortality	50 and over	both	26	12	x2.77	2025	x2.80
hiv.mortality	50 and over	male	26	15	x2.47	2025	x2.47
hiv.mortality	10-19	female	26	4	x2.27	2025	x1.21
hiv.mortality	15-24	female	26	23	x1.96	2025	x1.24
hiv.mortality	10-19	both	26	15	x1.76	2025	x0.90
prevalence	50 and over	male	26	0	x0.66	2025	x0.82
hiv.mortality	15-24	both	26	35	x1.63	2025	x1.02
engagement	0-14	both	11	0	+16.2 pts	2025	+20.4 pts
hiv.mortality	15-24	male	26	50	x1.57	2025	x0.68
suppression	0-14	both	7	0	+15.0 pts	2025	+17.5 pts
incidence	50 and over	male	26	19	x0.70	2025	x1.06
incidence	10-19	male	26	15	x1.39	2025	x1.32
awareness	0-14	both	11	0	+10.9 pts	2025	+13.3 pts

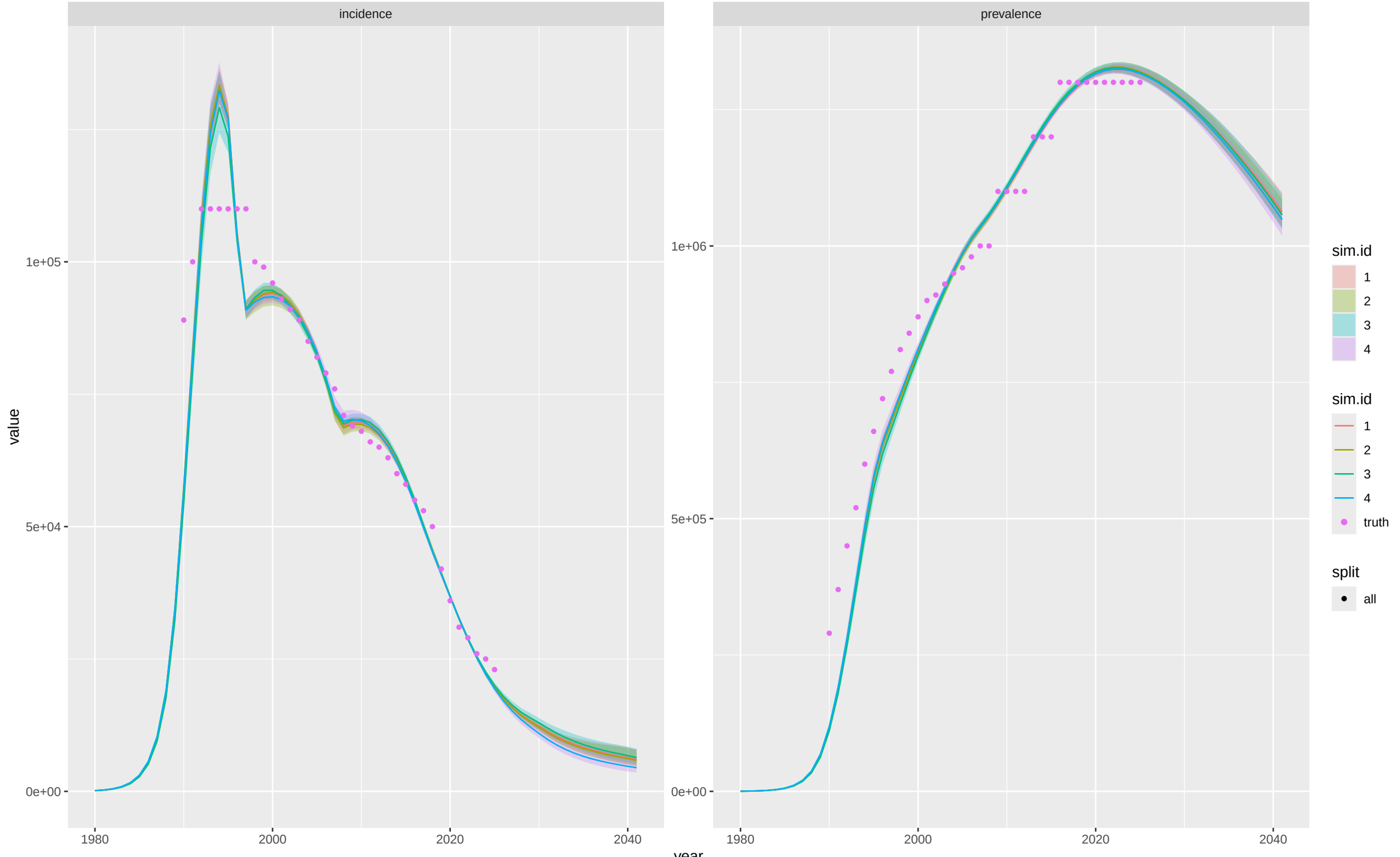
Zambia step 2: likelihood



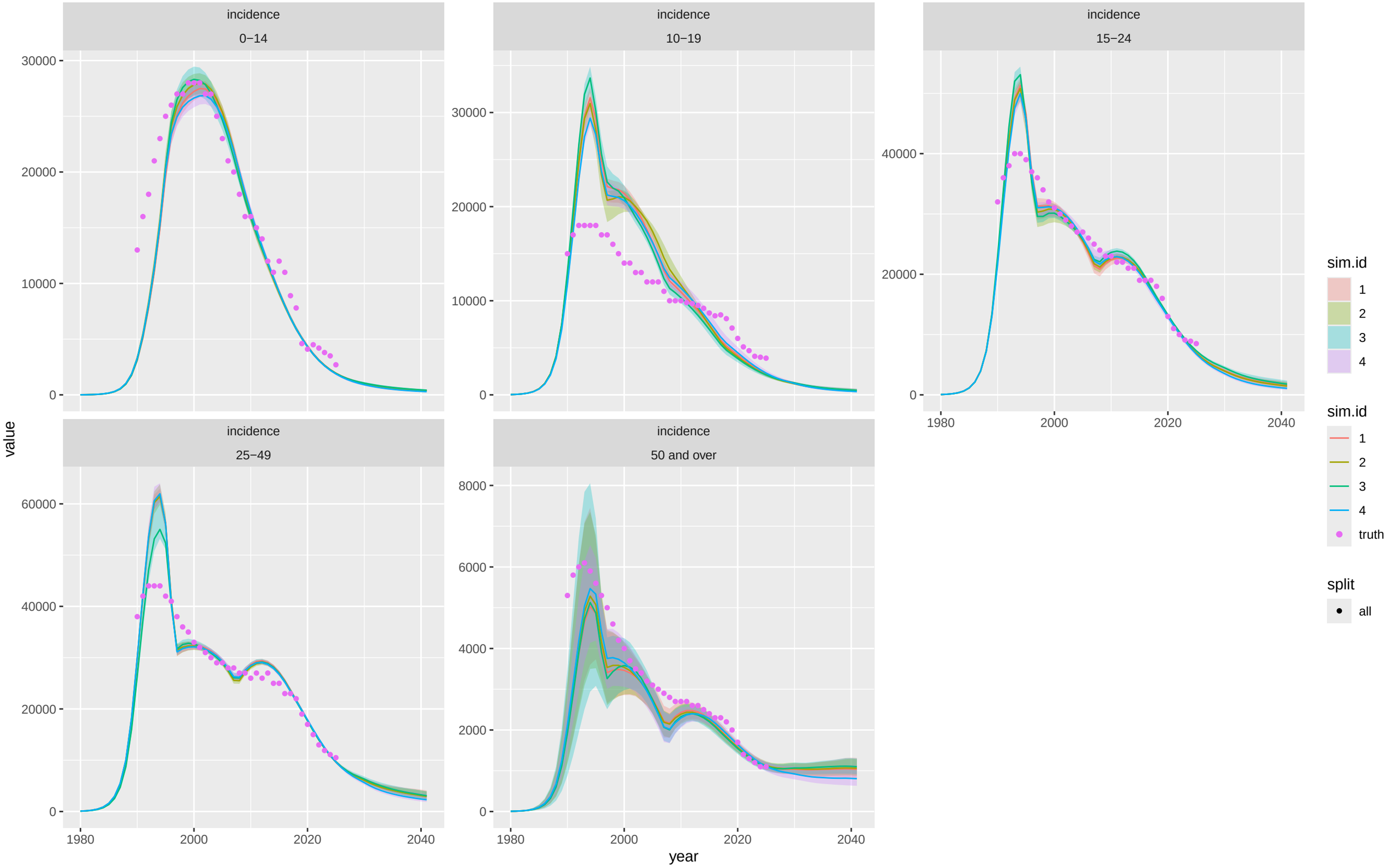
Zambia step 2: trace



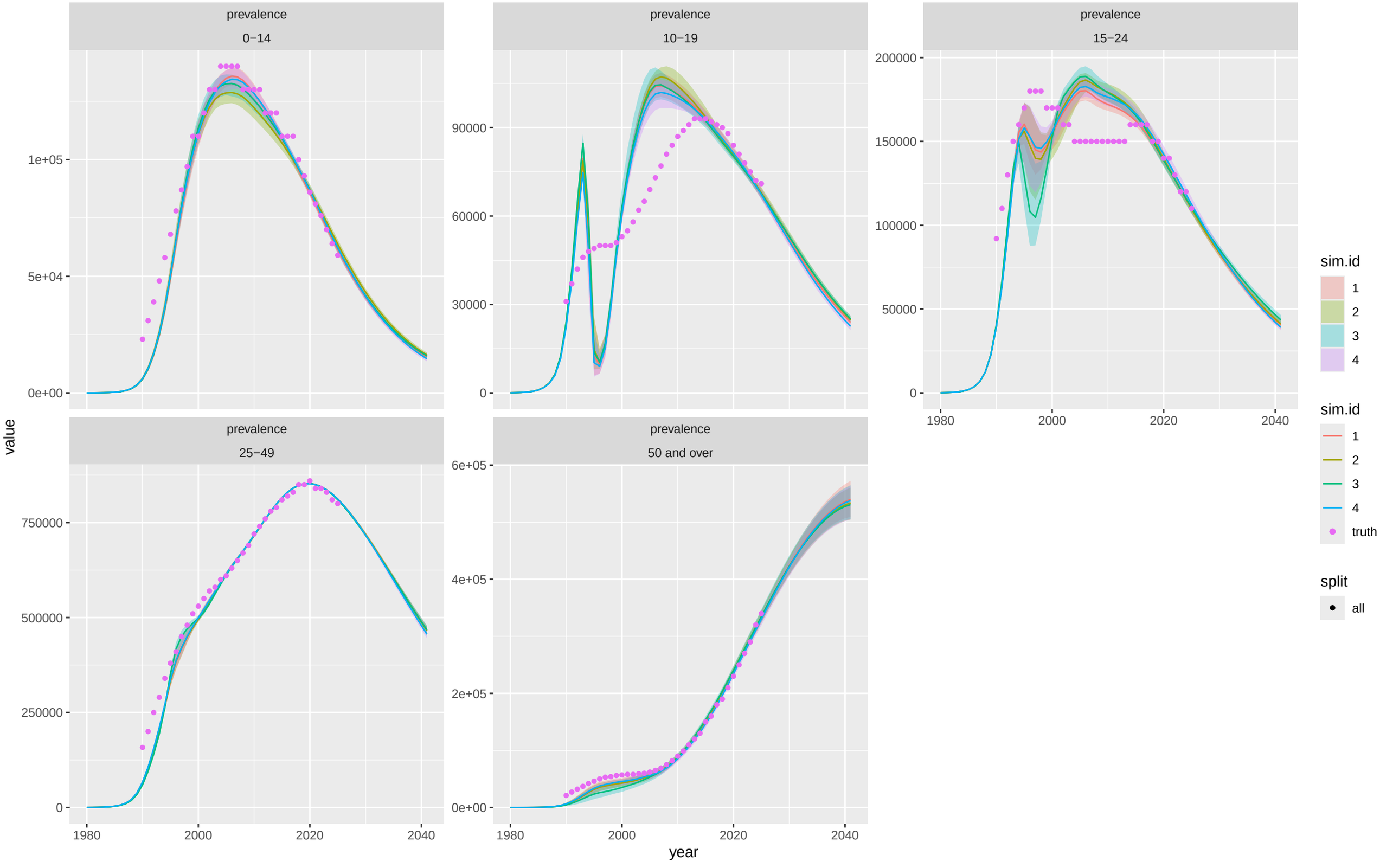
Zambia step 2: fit (one color per chain)



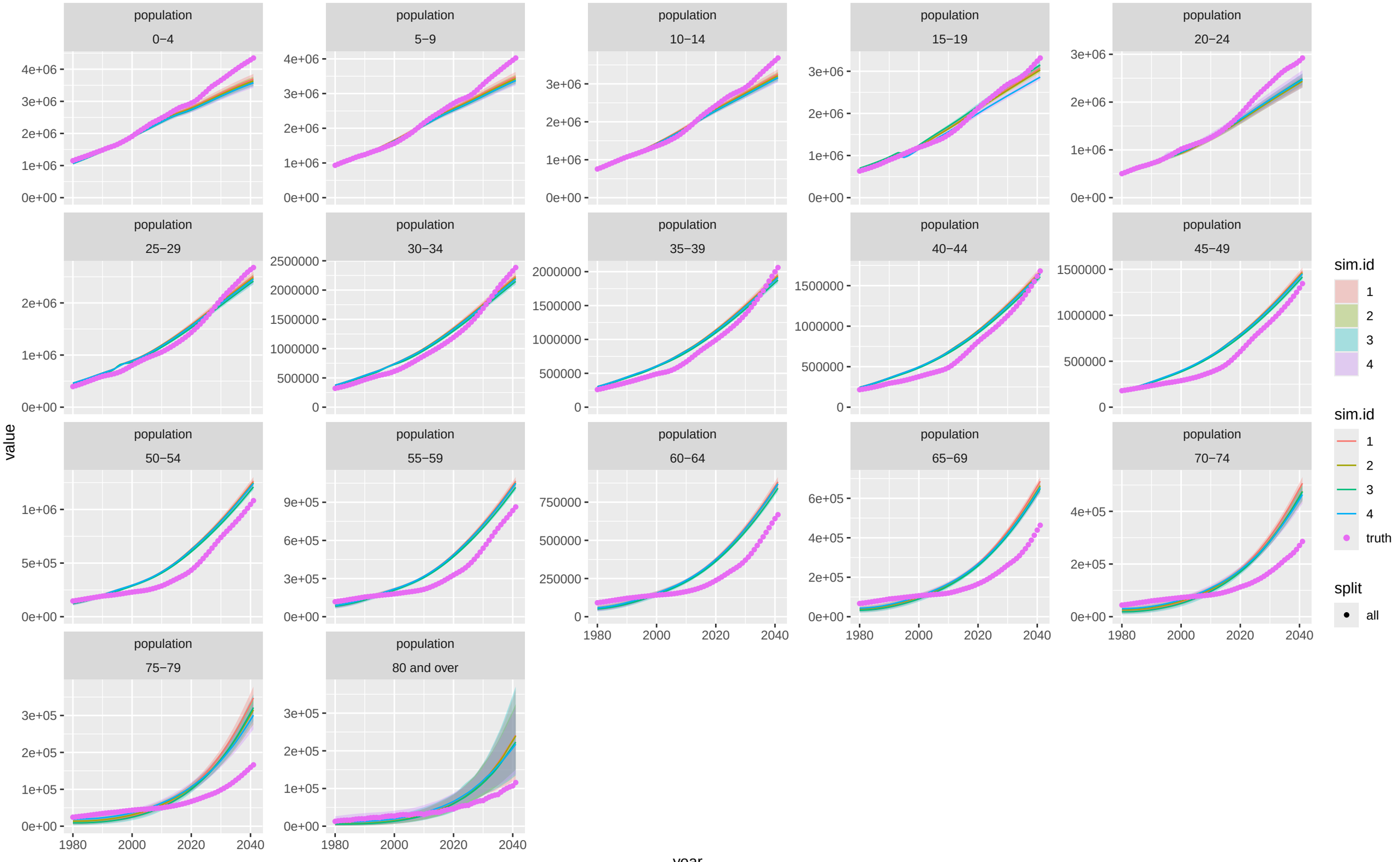
Zambia step 2: incidence.by.age (one color per chain)



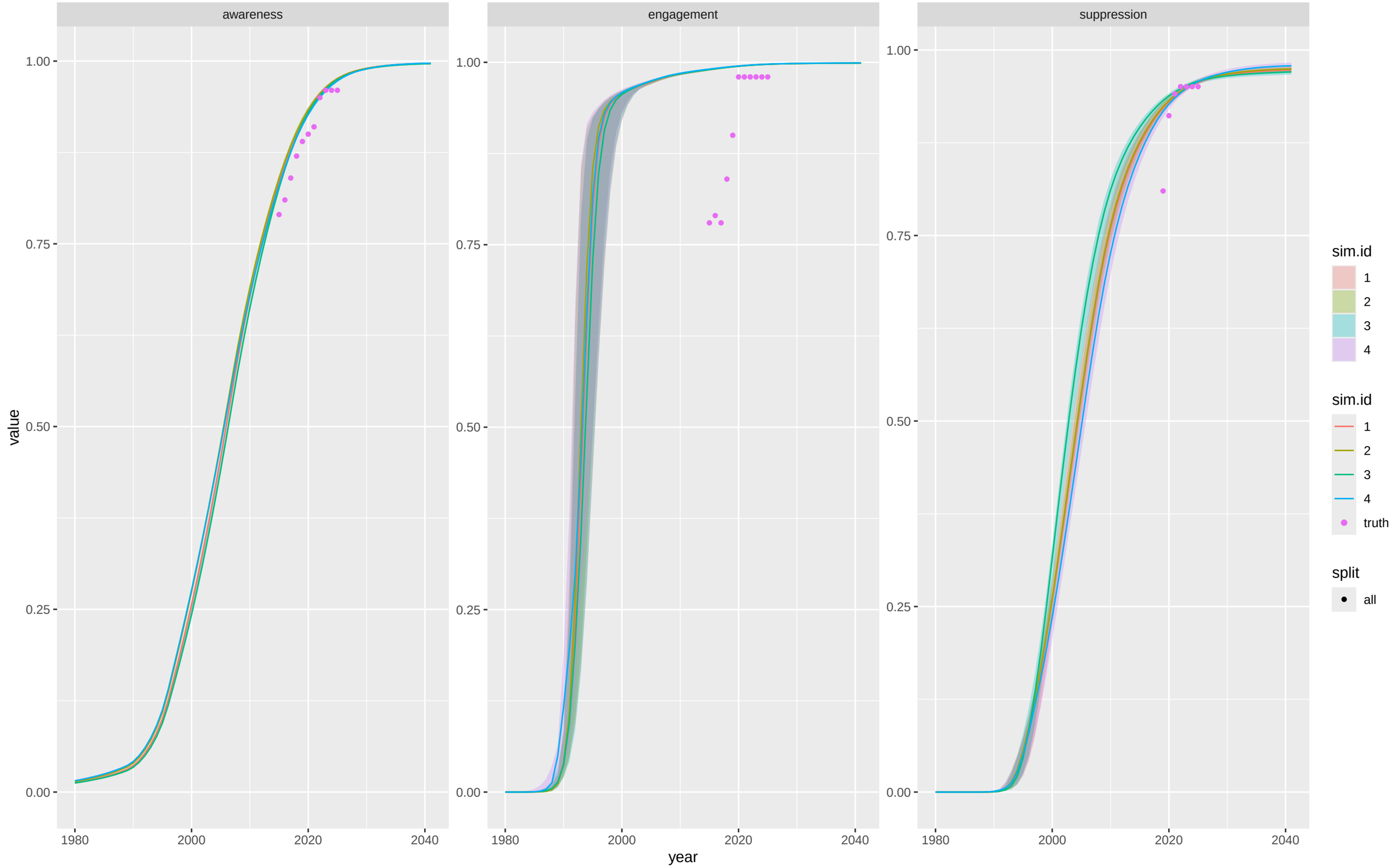
Zambia step 2: prevalence.by.age (one color per chain)



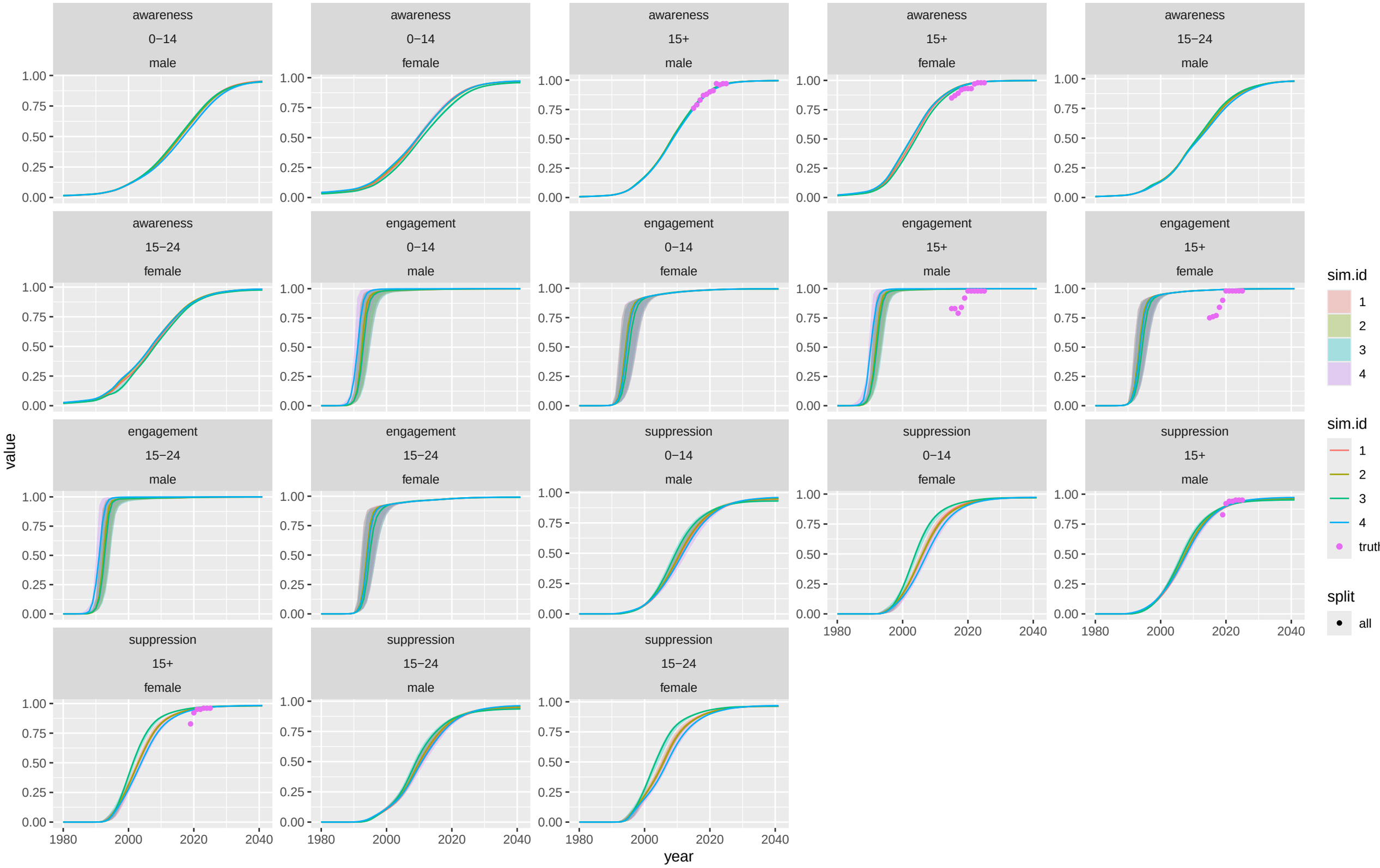
Zambia step 2: population.by.age (one color per chain)



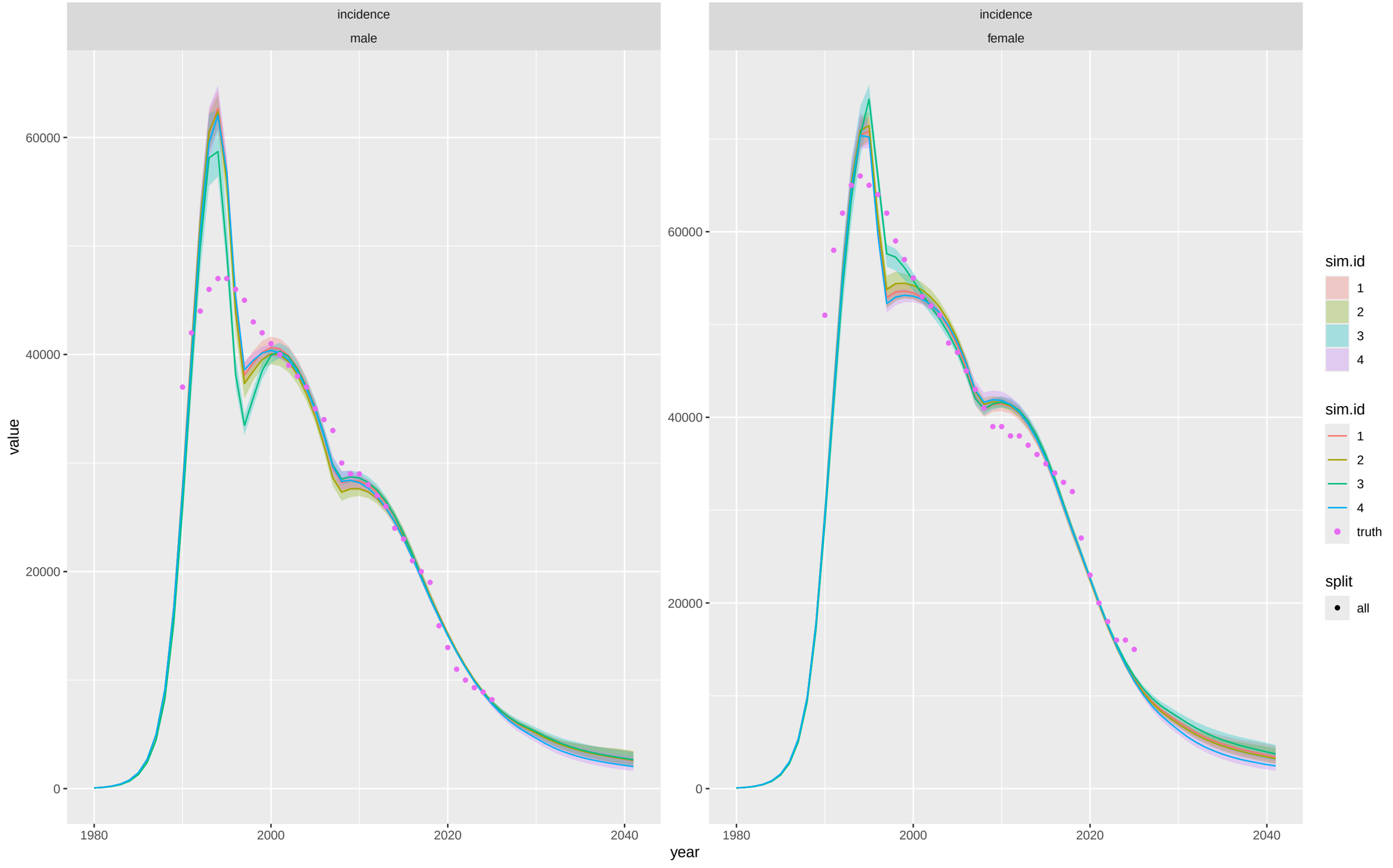
Zambia step 2: cascade (one color per chain)



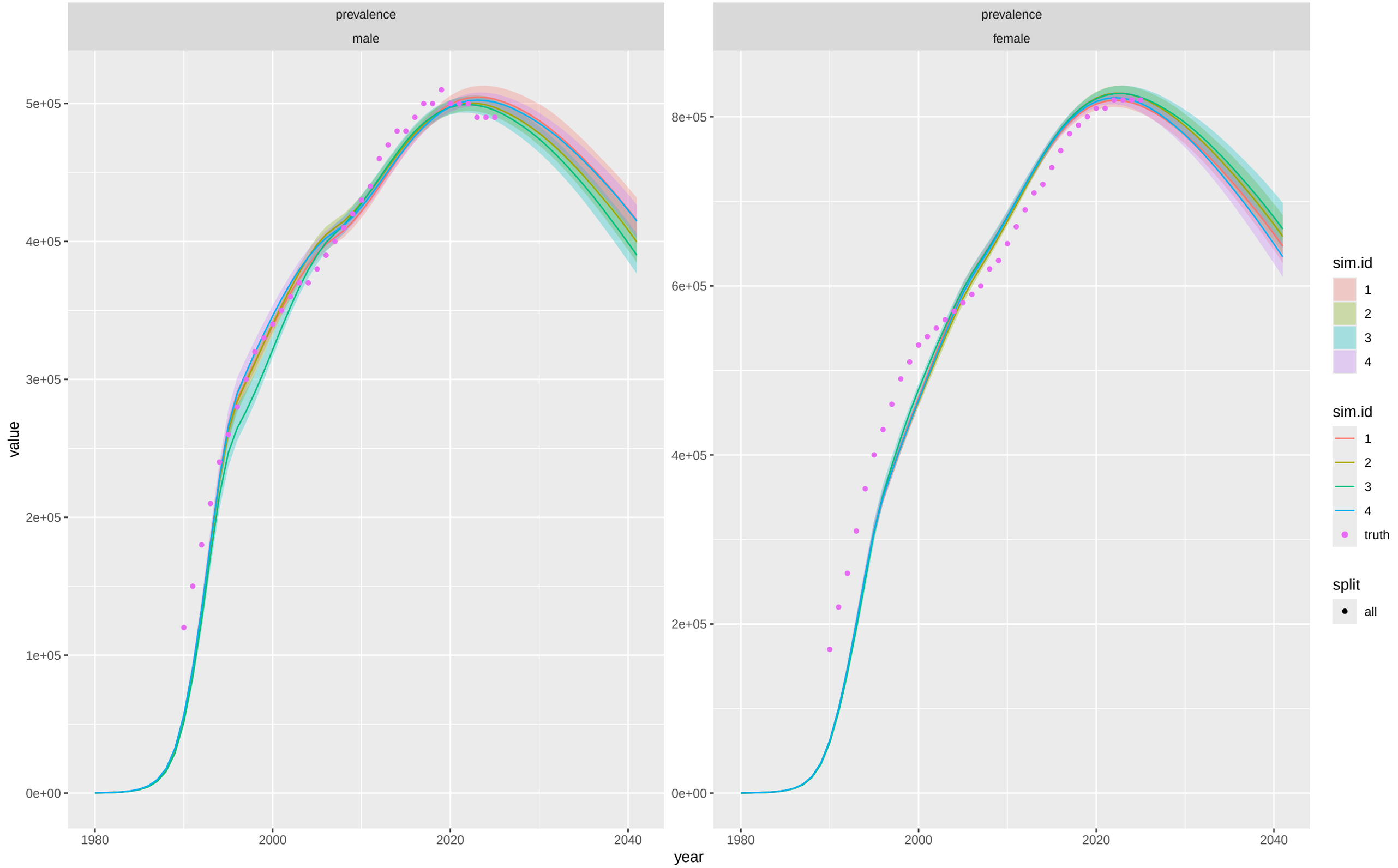
Zambia step 2: cascade.by.age.sex (one color per chain)



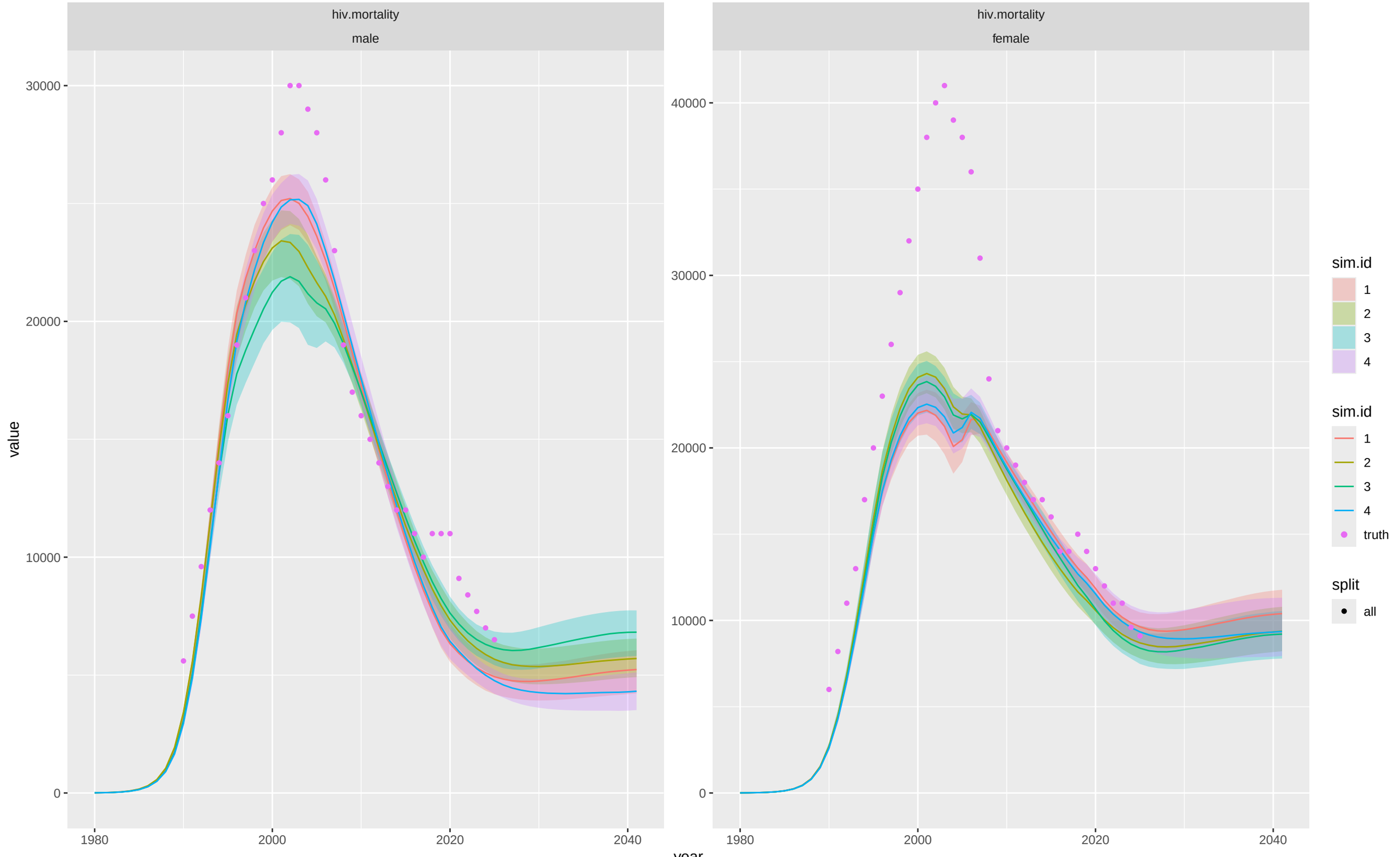
Zambia step 2: incidence.by.sex (one color per chain)



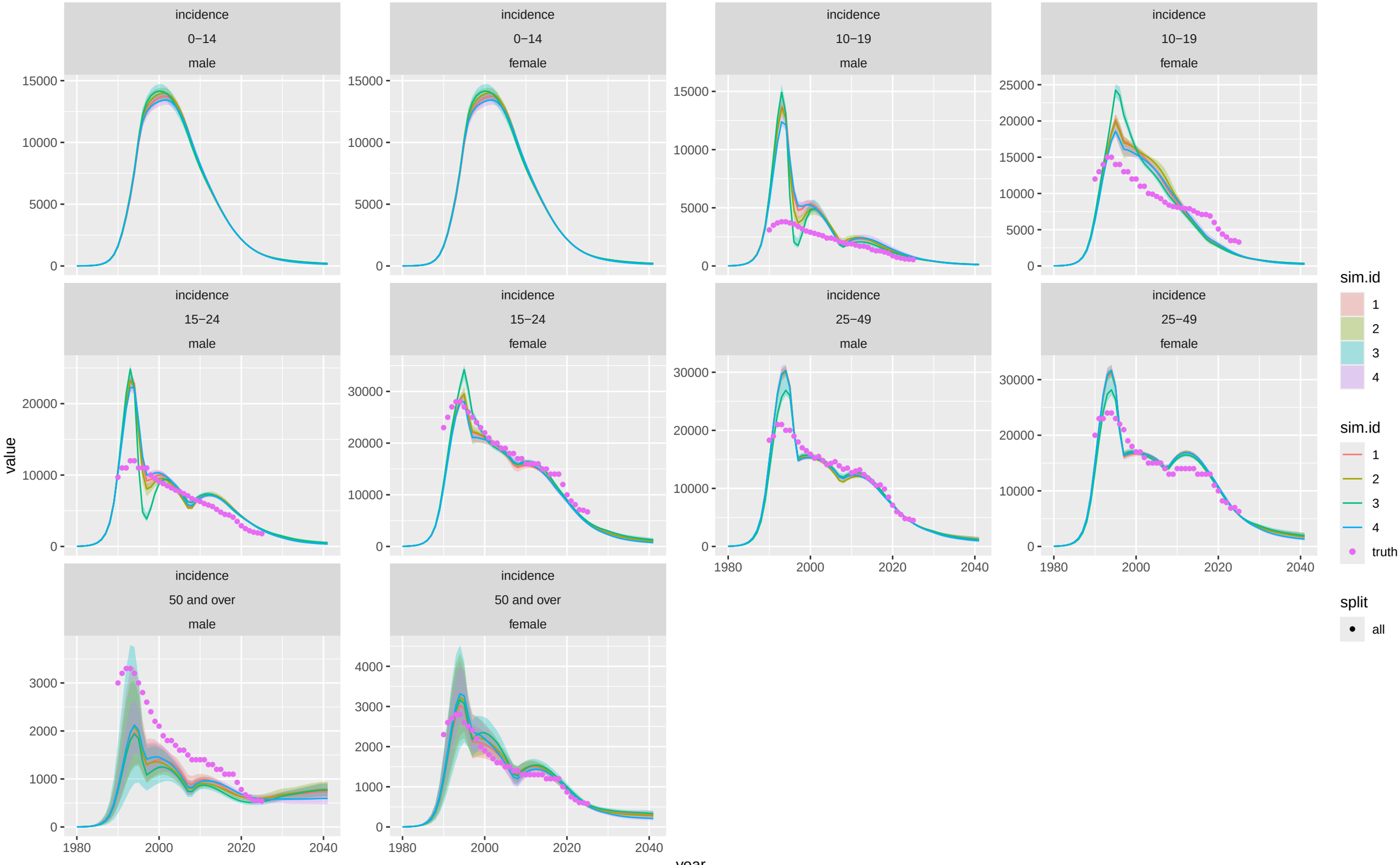
Zambia step 2: prevalence.by.sex (one color per chain)



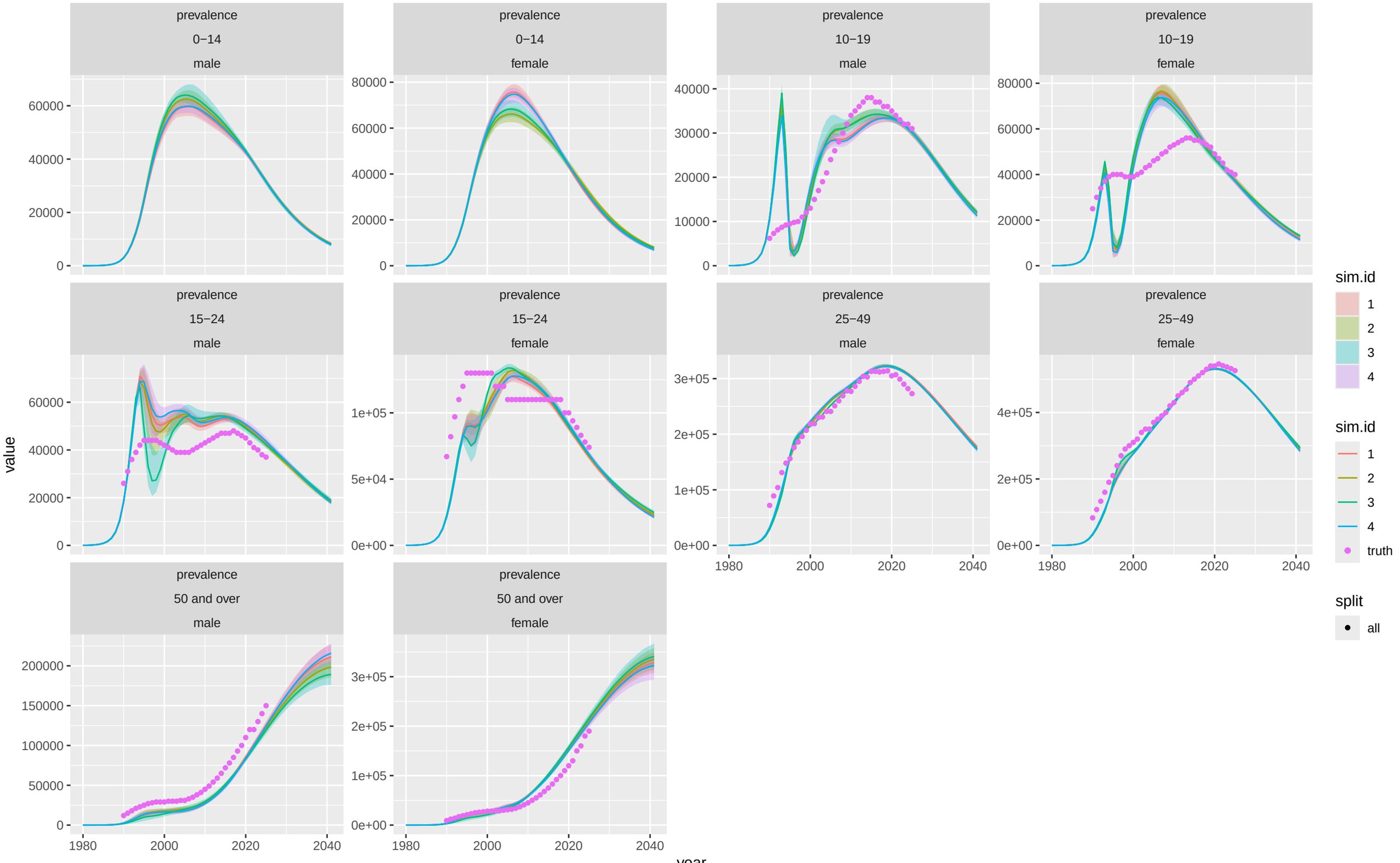
Zambia step 2: hiv.mortality.by.sex (one color per chain)



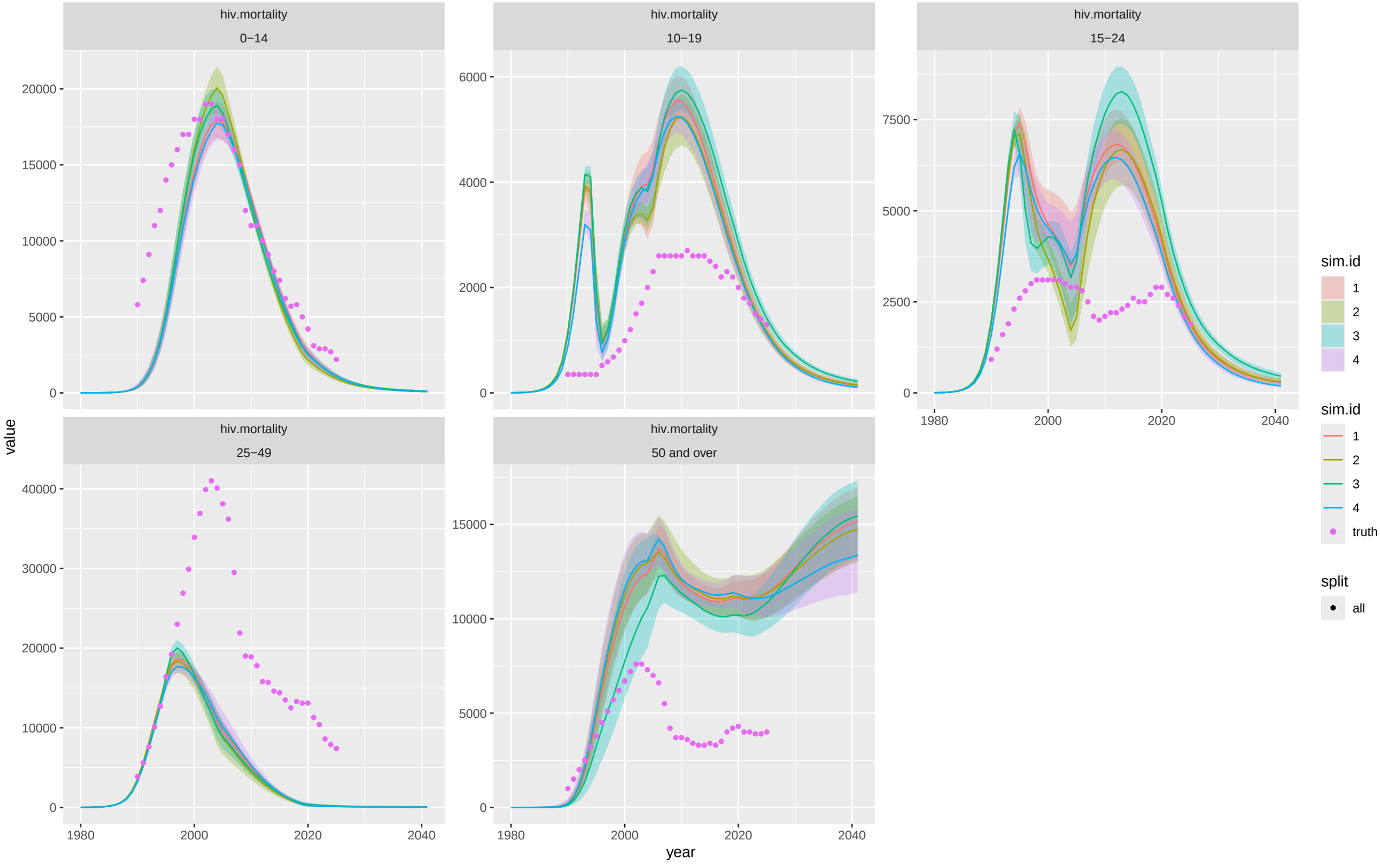
Zambia step 2: incidence.by.age.sex (one color per chain)



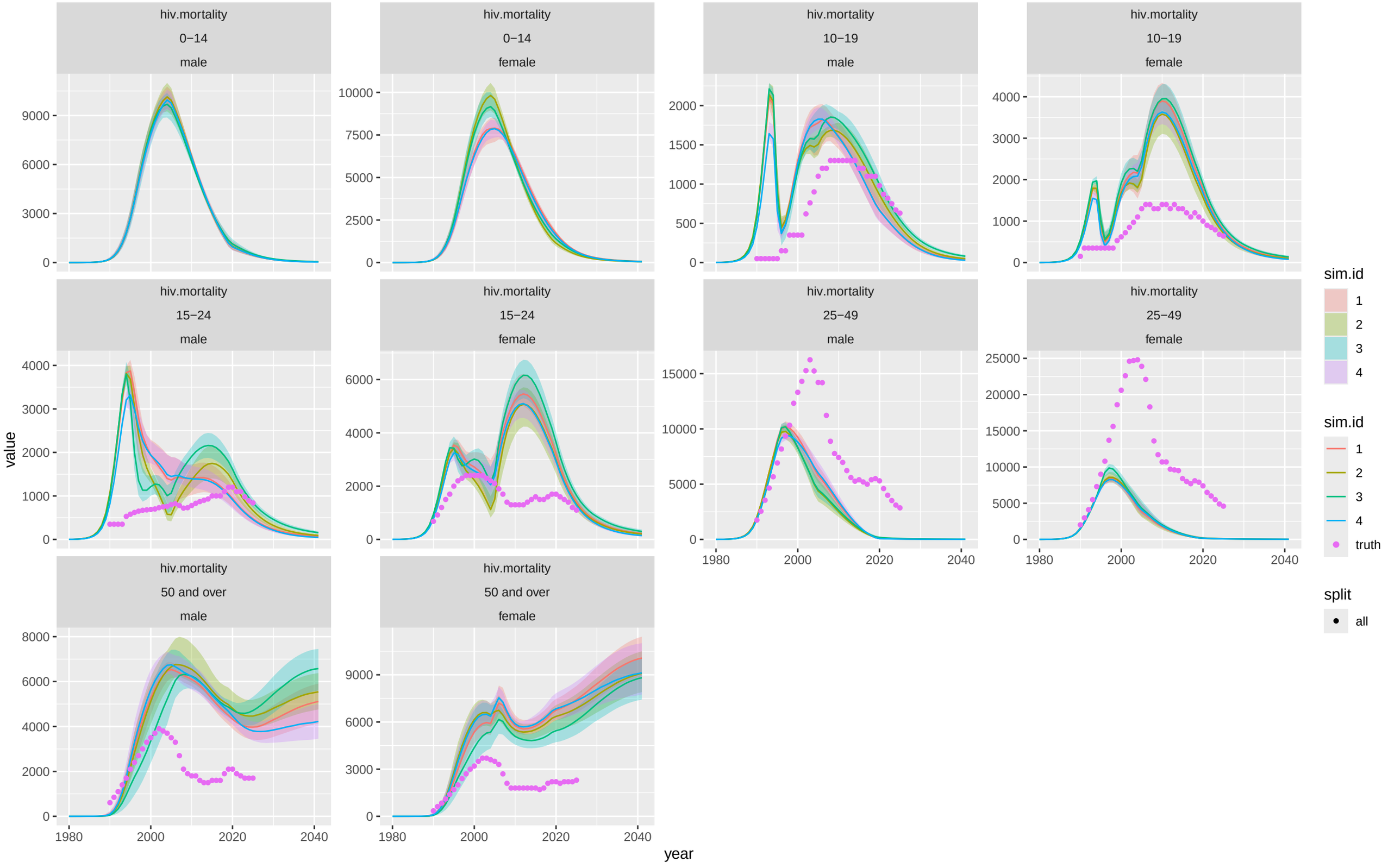
Zambia step 2: prevalence.by.age.sex (one color per chain)



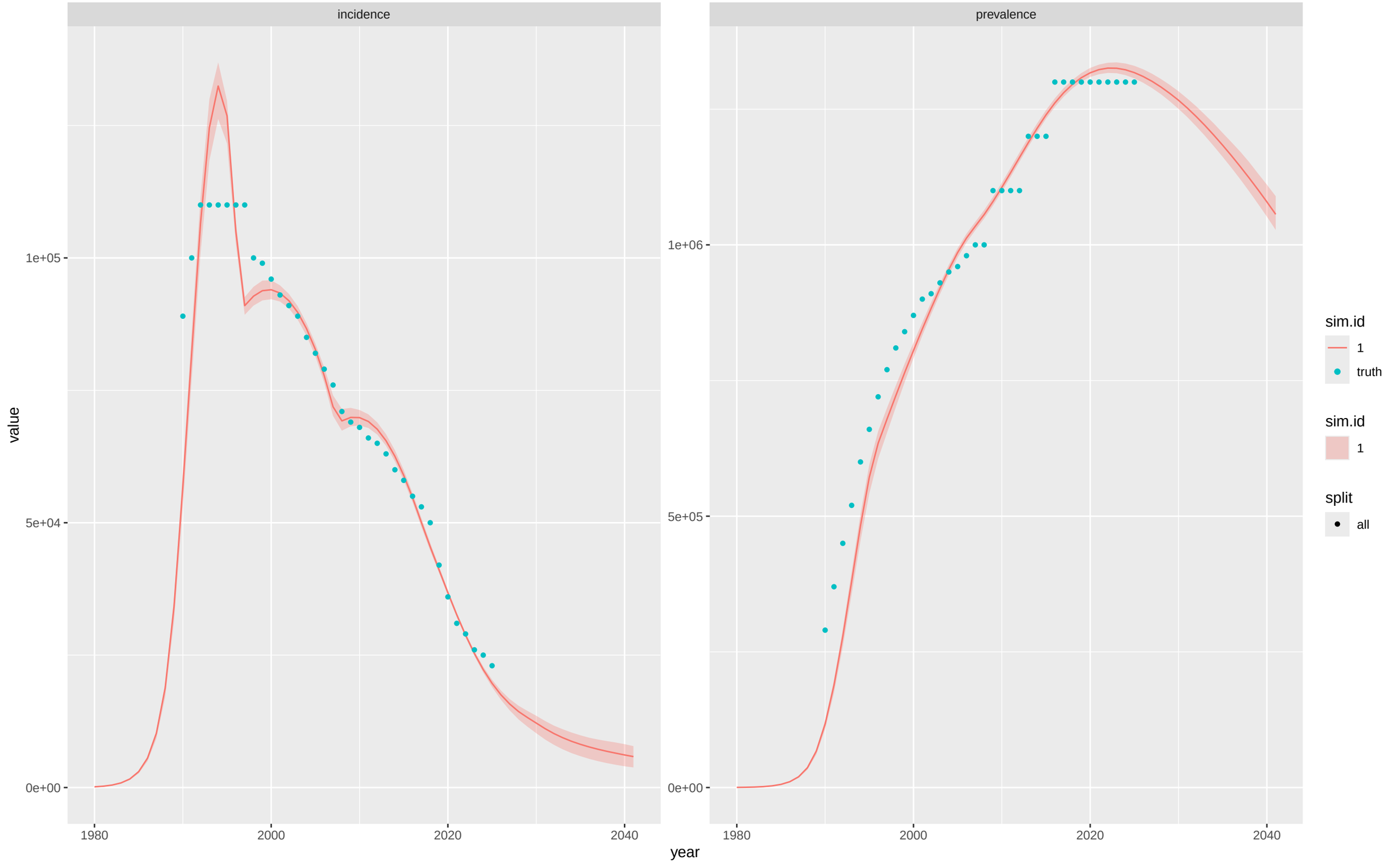
Zambia step 2: hiv.mortality.by.age (one color per chain)



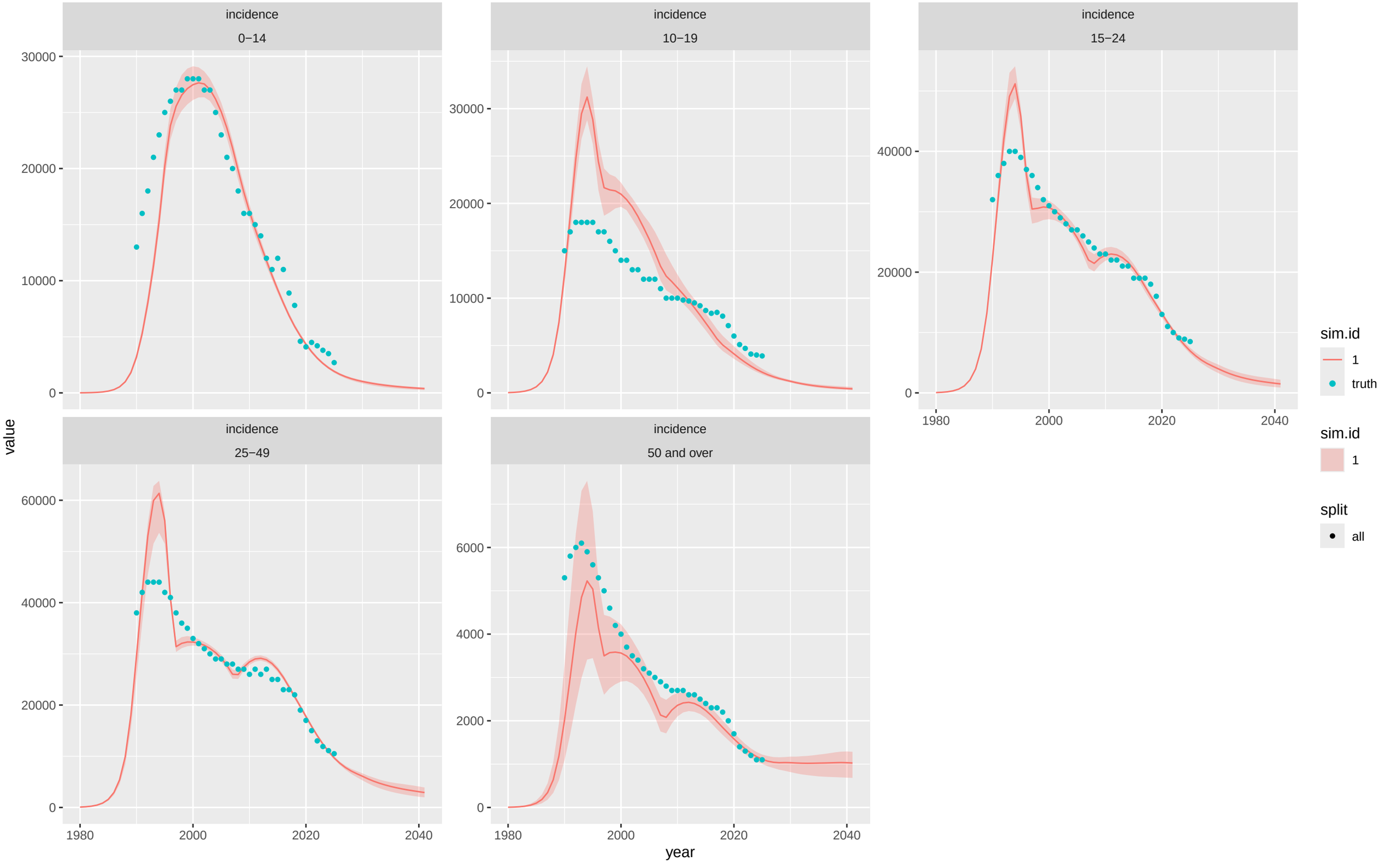
Zambia step 2: hiv.mortality.by.age.sex (one color per chain)



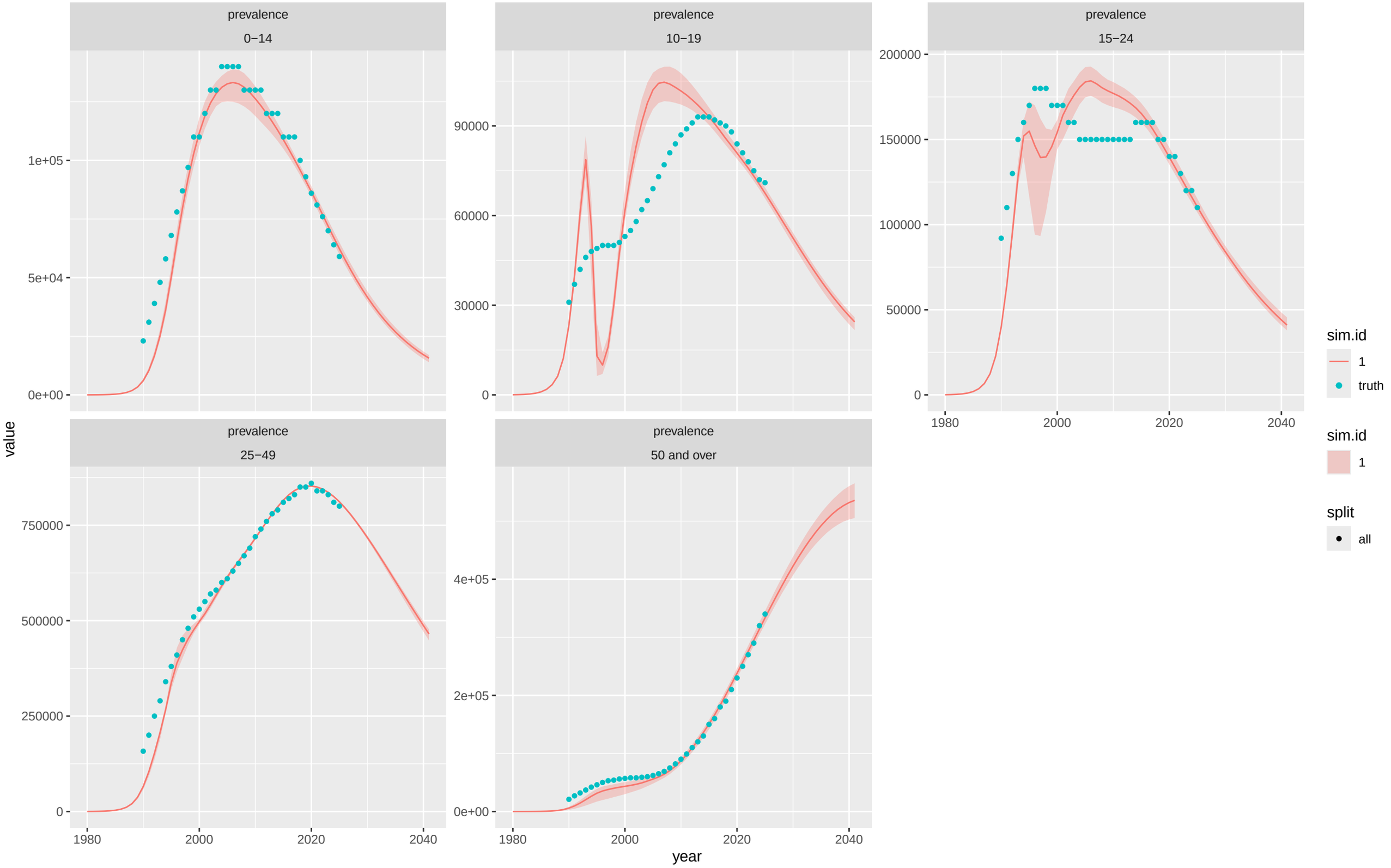
Zambia: fit (merged Step 2 simset)



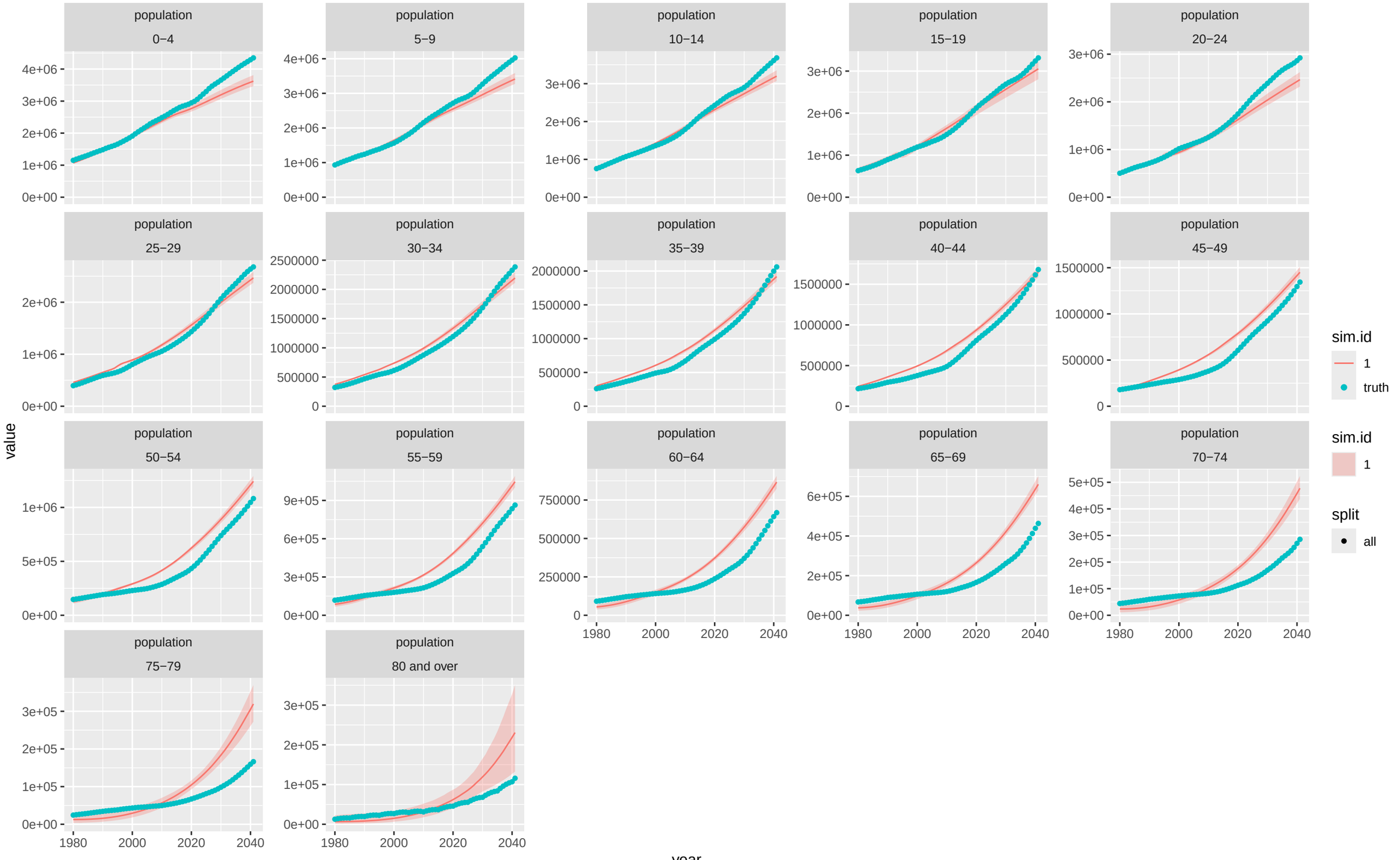
Zambia: incidence.by.age (merged Step 2 simset)



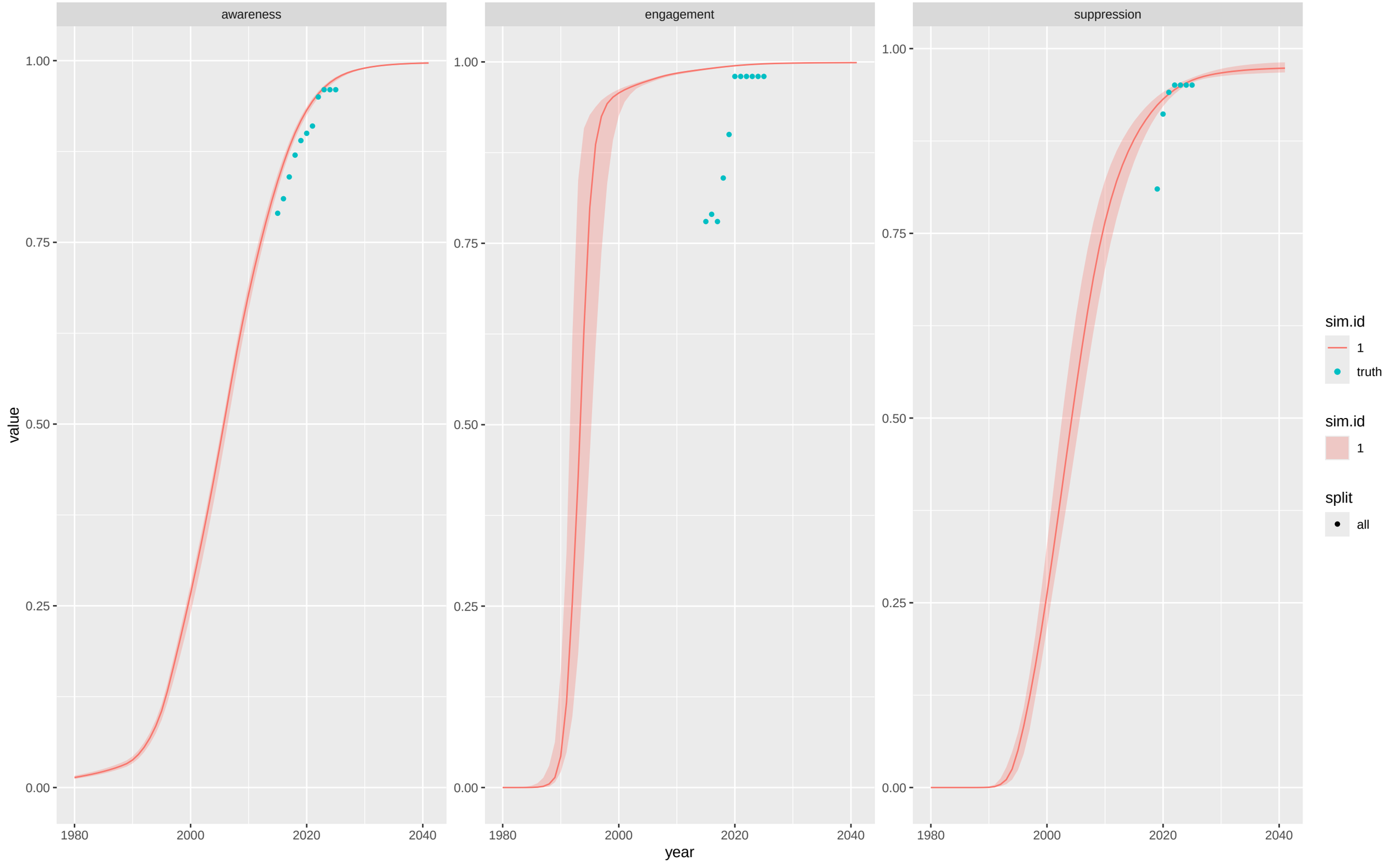
Zambia: prevalence.by.age (merged Step 2 simset)



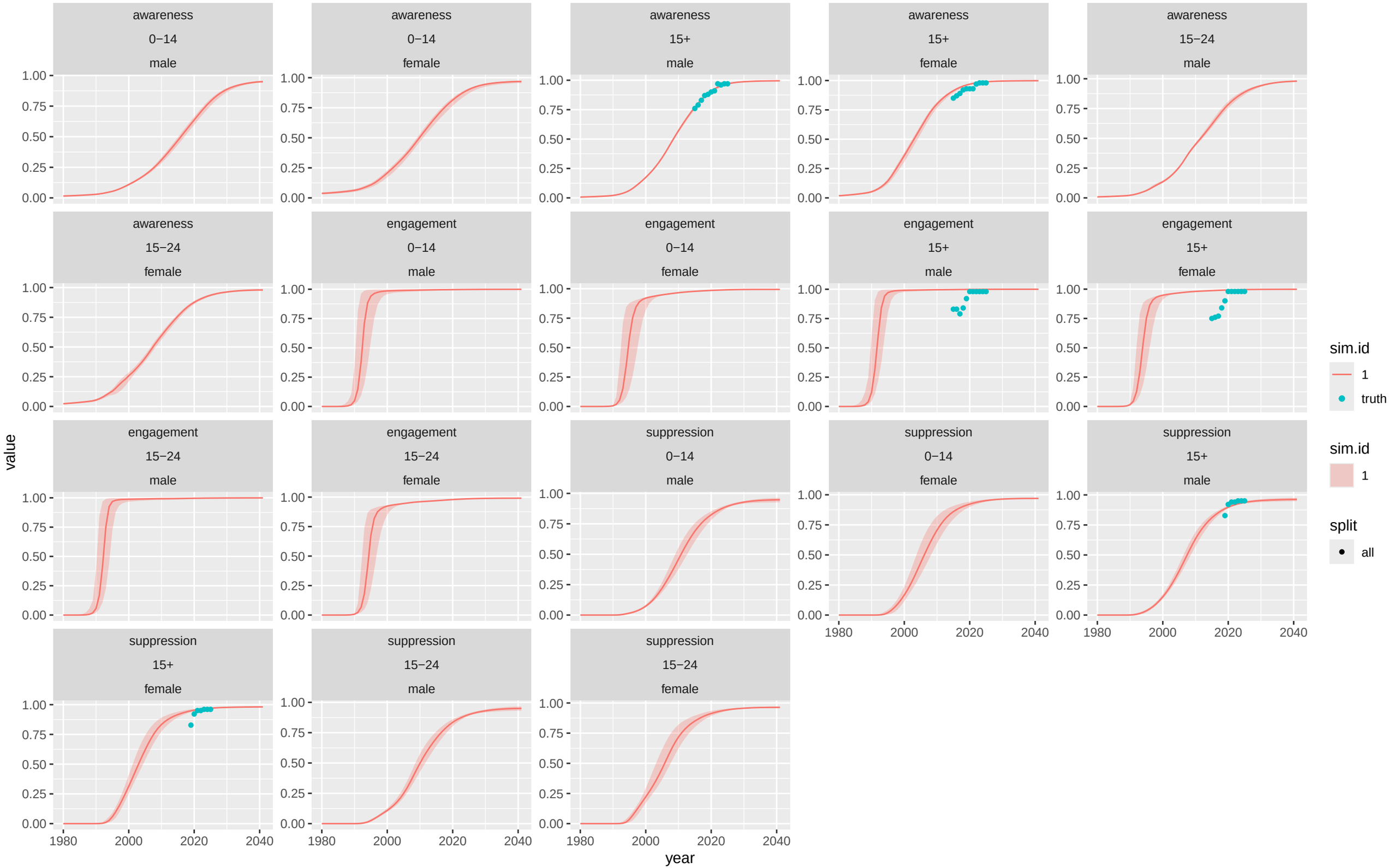
Zambia: population.by.age (merged Step 2 simset)



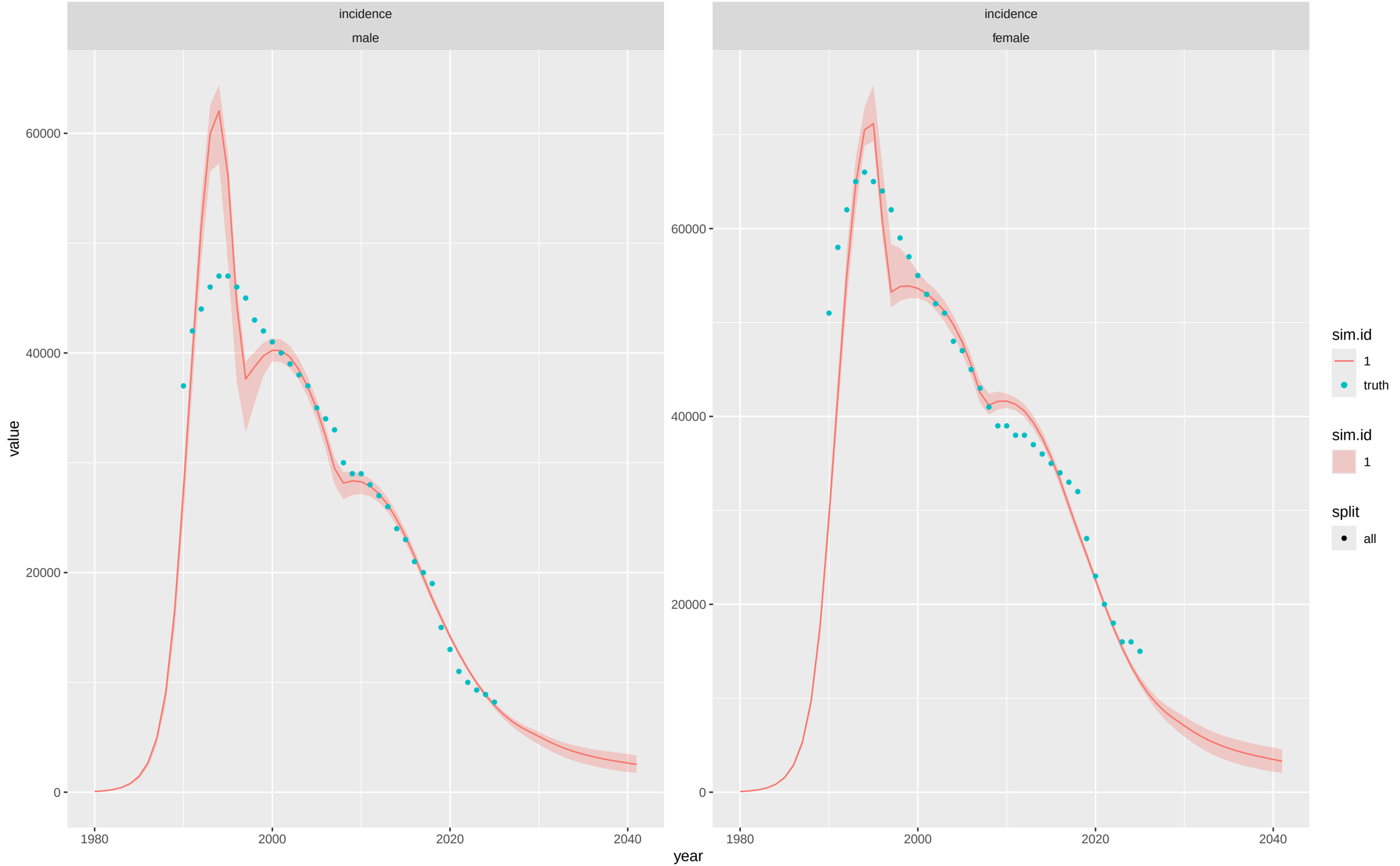
Zambia: cascade (merged Step 2 simset)



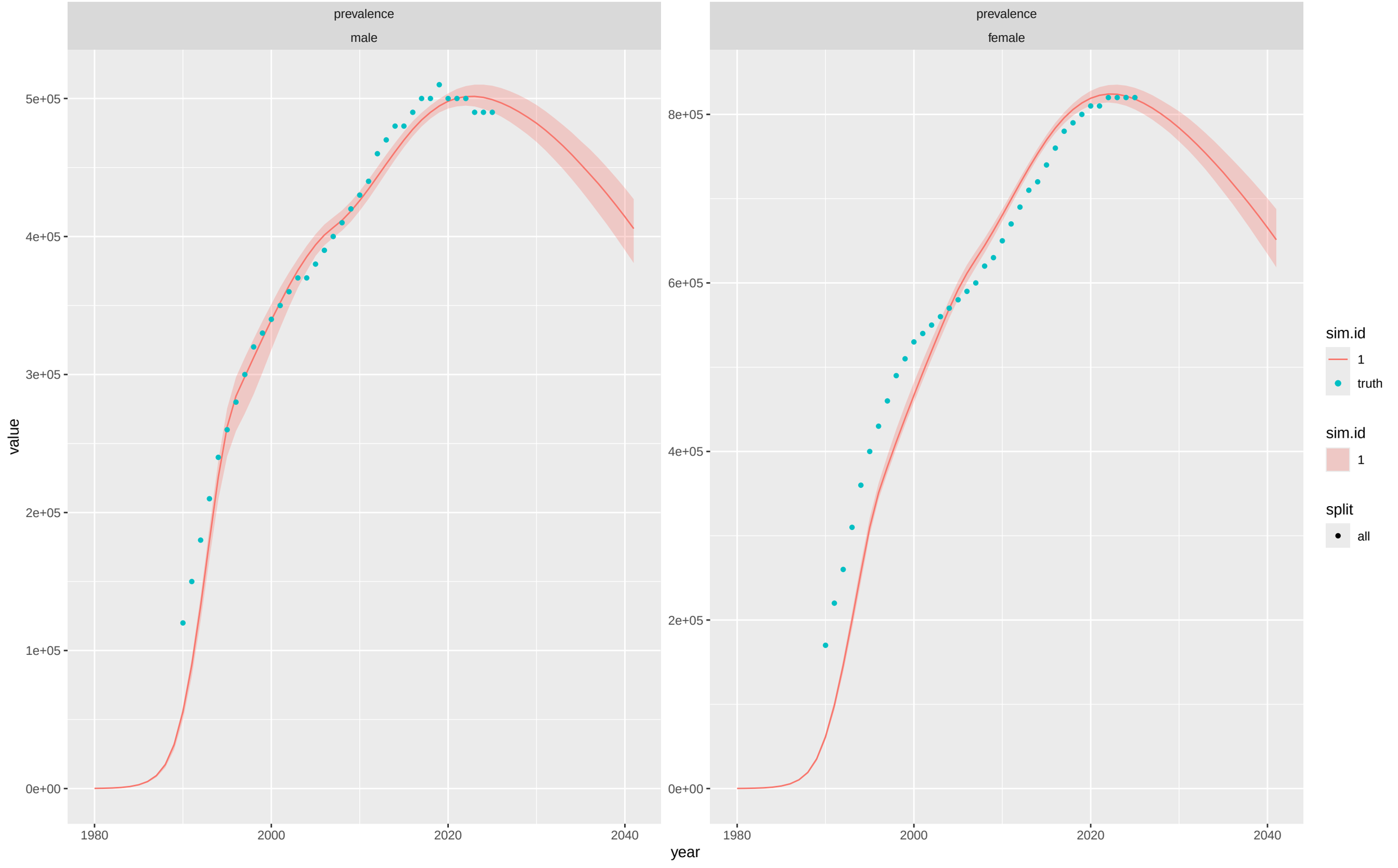
Zambia: cascade.by.age.sex (merged Step 2 simset)



Zambia: incidence.by.sex (merged Step 2 simset)



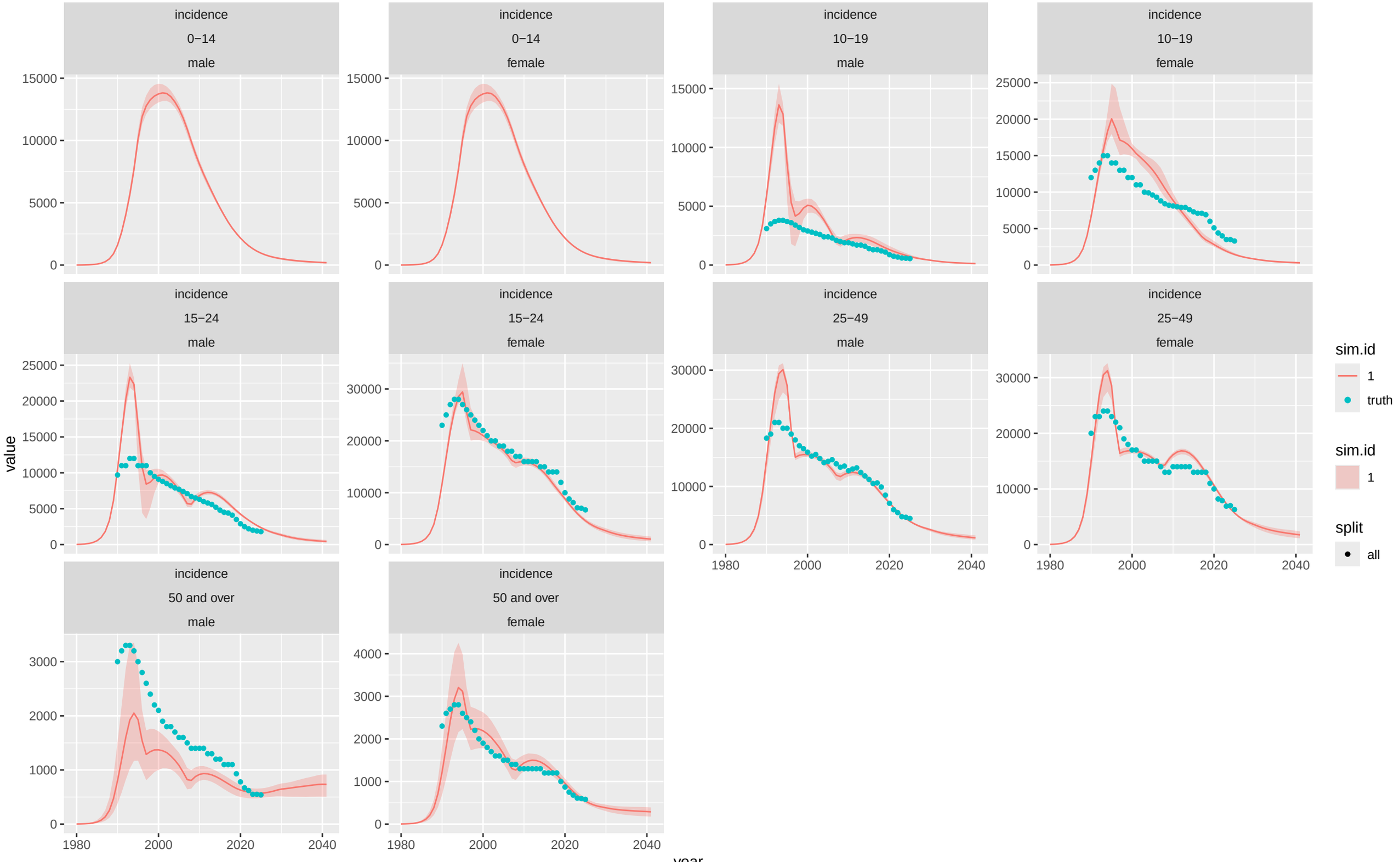
Zambia: prevalence.by.sex (merged Step 2 simset)



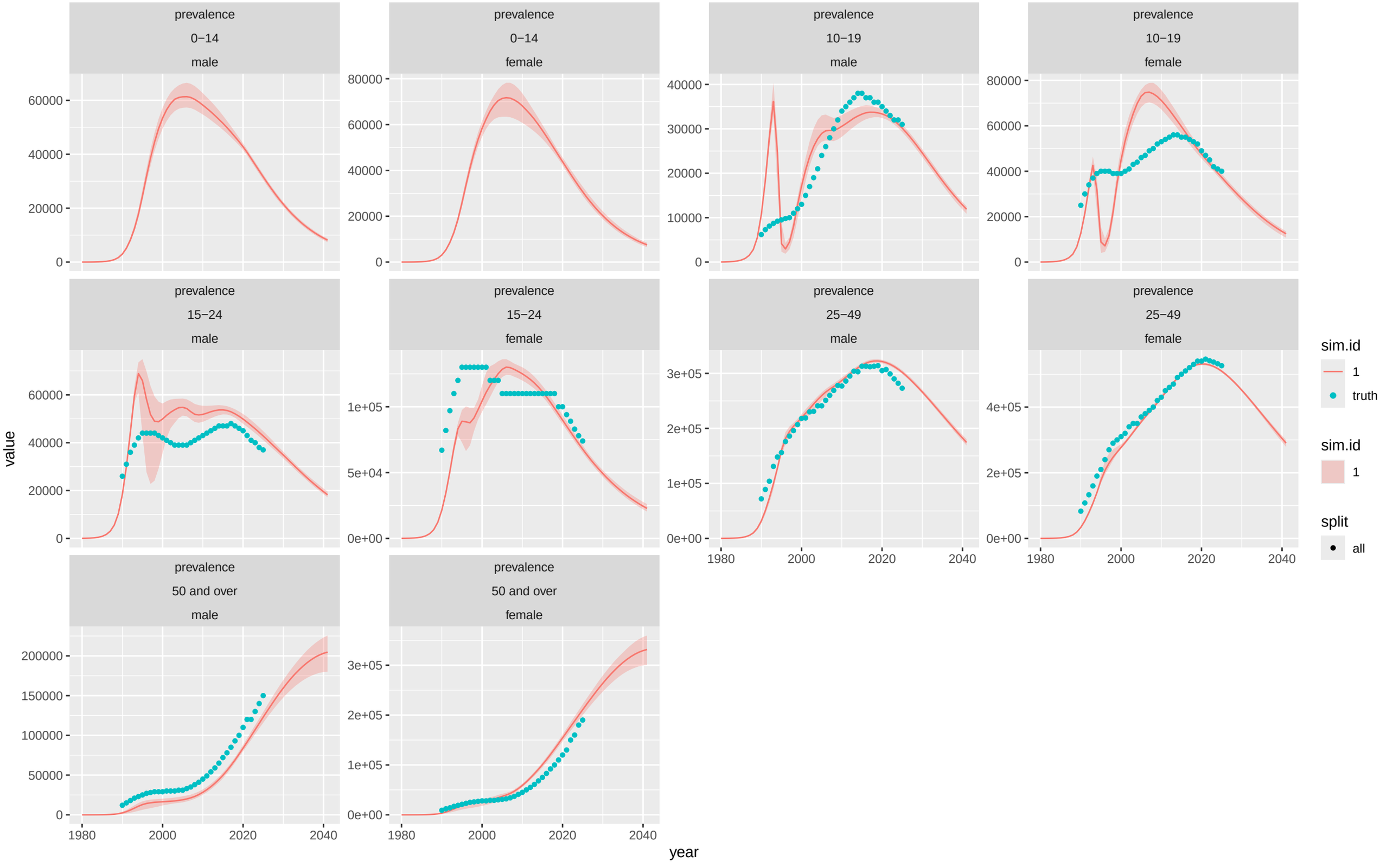
Zambia: hiv.mortality.by.sex (merged Step 2 simset)



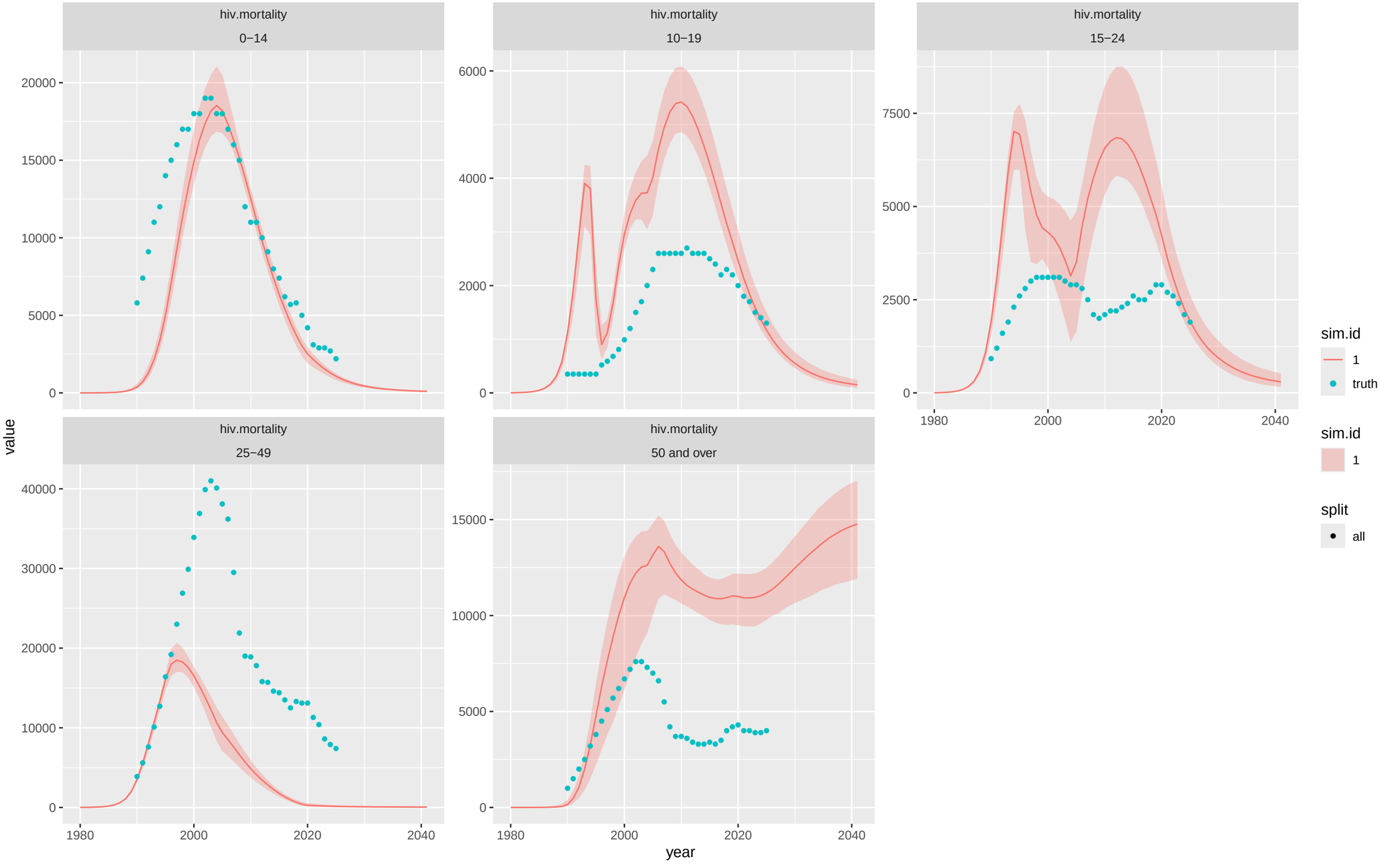
Zambia: incidence.by.age.sex (merged Step 2 simset)



Zambia: prevalence.by.age.sex (merged Step 2 simset)



Zambia: hiv.mortality.by.age (merged Step 2 simset)



Zambia: hiv.mortality.by.age.sex (merged Step 2 simset)

